Heart Disease Prediction Using Machine Learning

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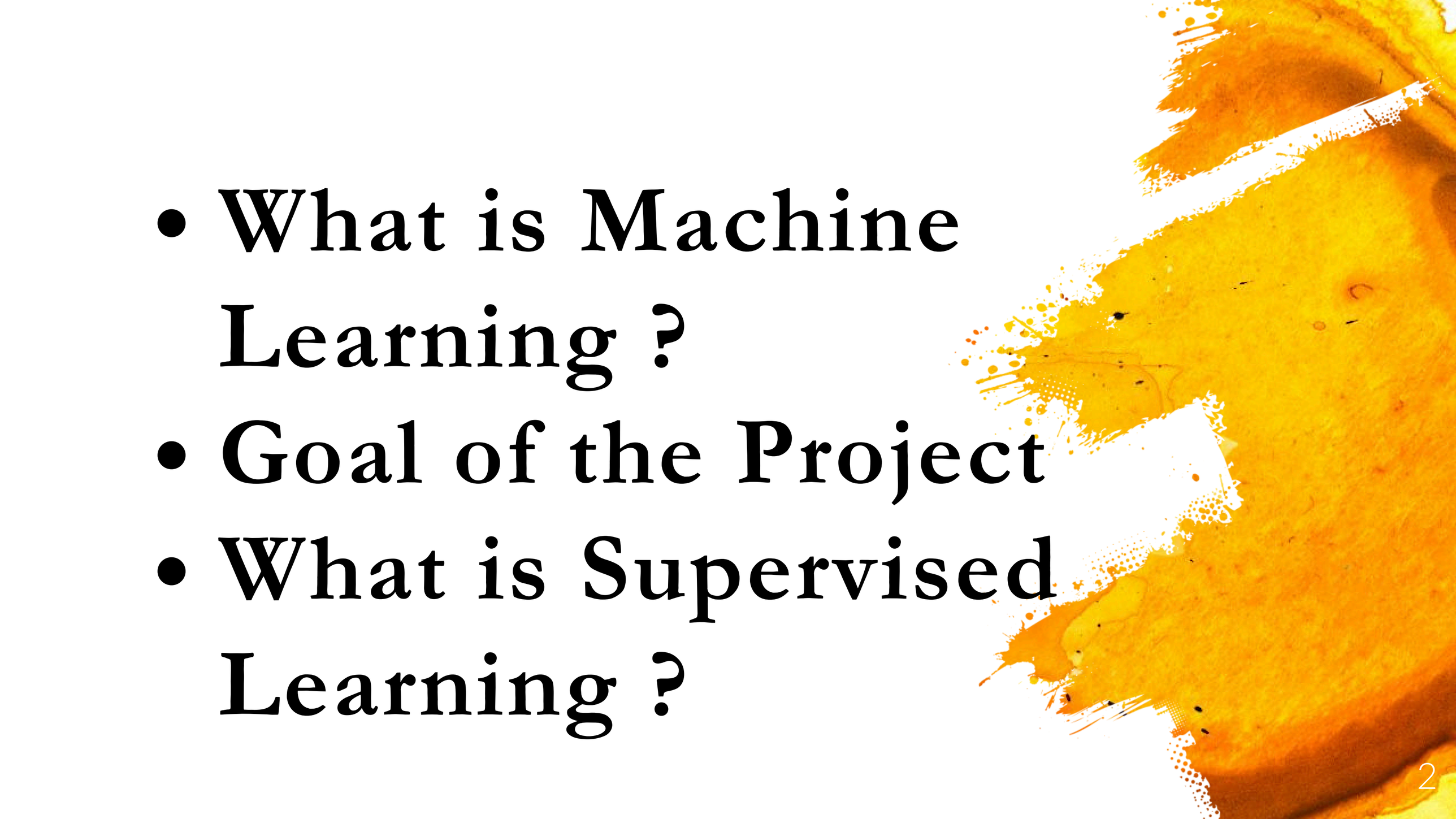
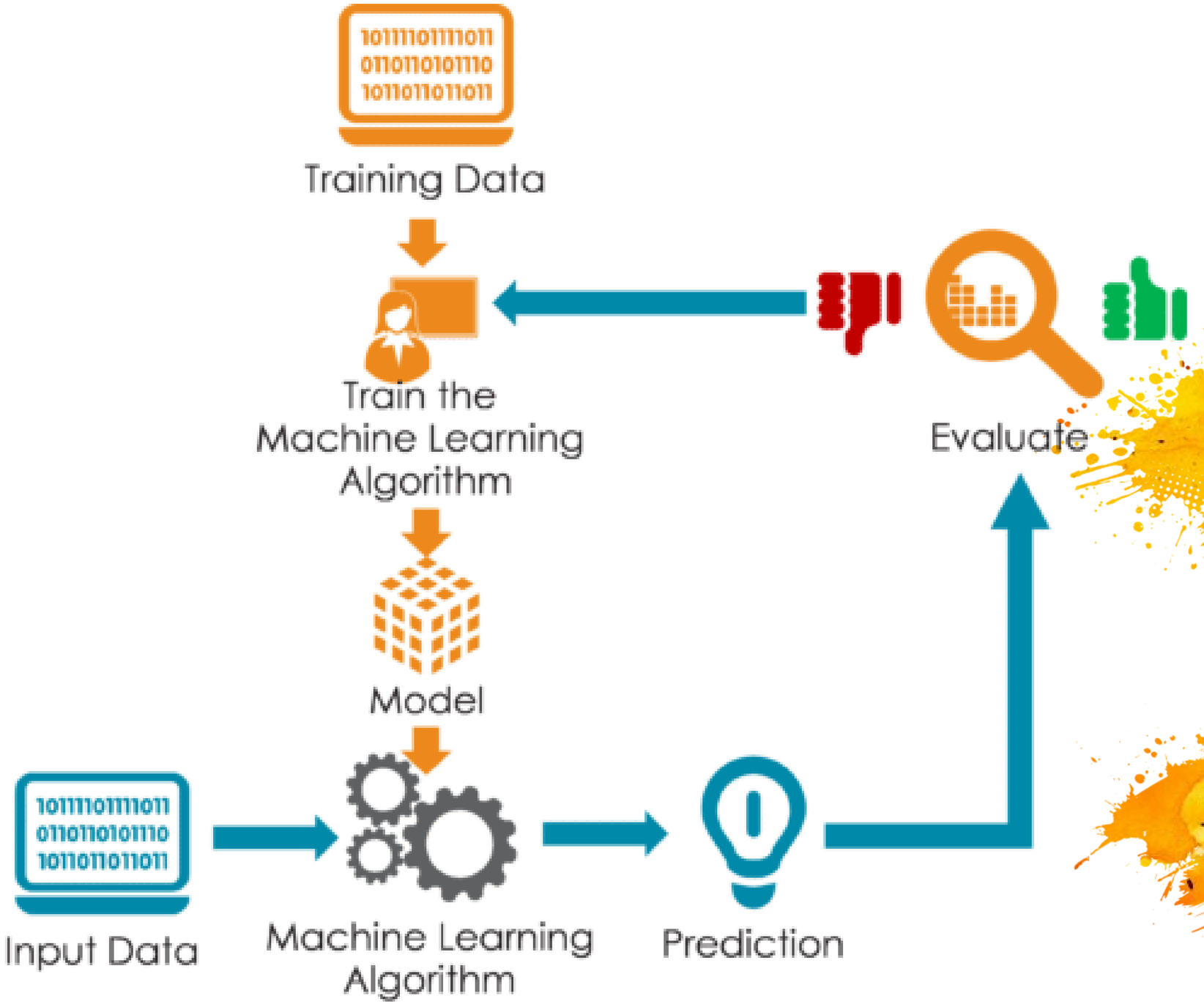
- What is Machine Learning ?
- Goal of the Project
- What is Supervised Learning ?

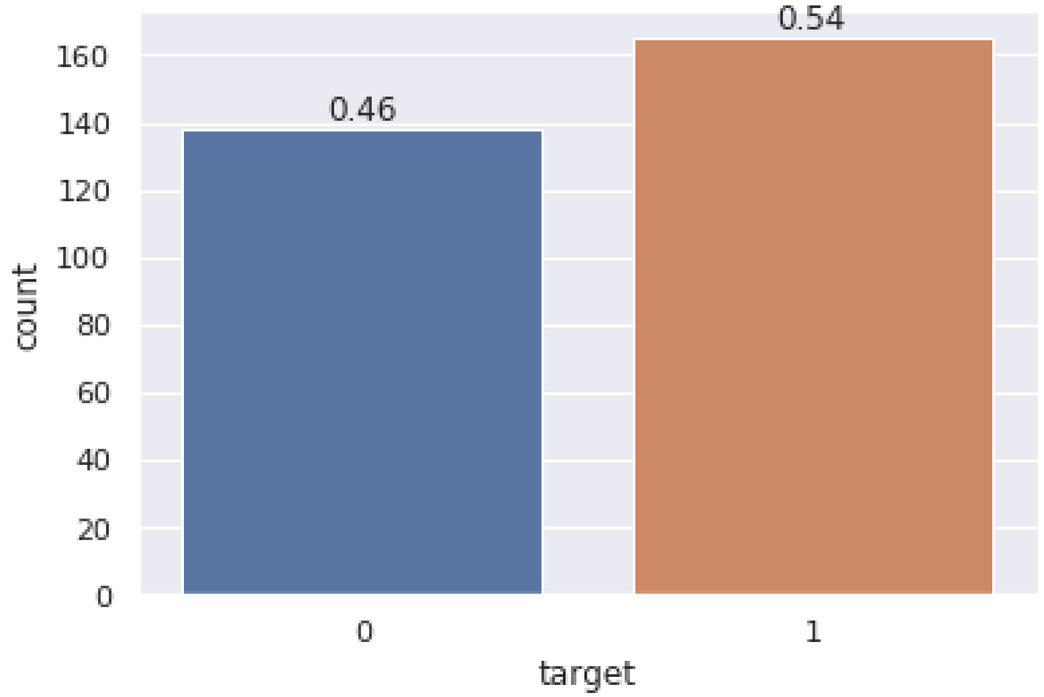



Training Data
Train the Machine Learning Algorithm
Evaluate
Model
Input Data
Machine Learning Algorithm
Prediction

of the Project

Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
2	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
3	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
4	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
5	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
6	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
7	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
8	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
9	44	1	1	120	263	0	1	173	0	0	2	0	3	1
10	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
11	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1
12	54	1	0	140	239	0	1	160	0	1.2	2	0	2	1
13	48	0	2	130	275	0	1	139	0	0.2	2	0	2	1
14	49	1	1	130	266	0	1	171	0	0.6	2	0	2	1



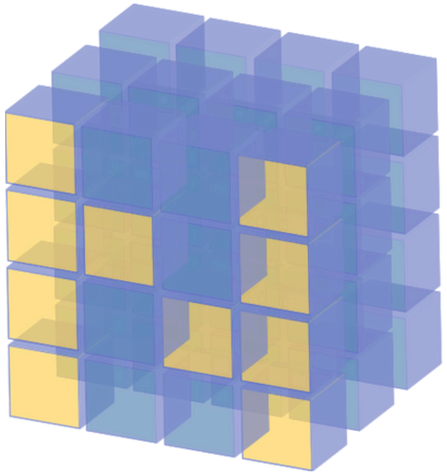

```
1      165
0      138
```

3 step process

Split the dataset

Train the dataset

Compare the Algos

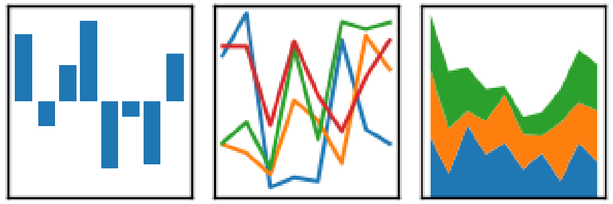


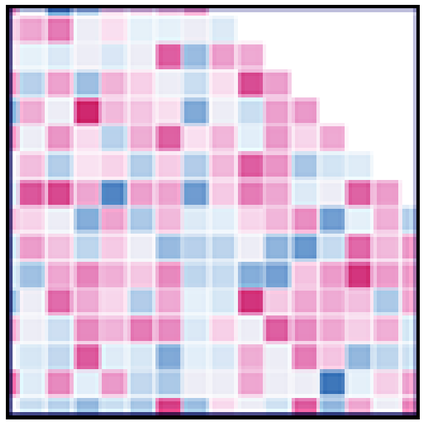
scikit learn

NumPy

pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$

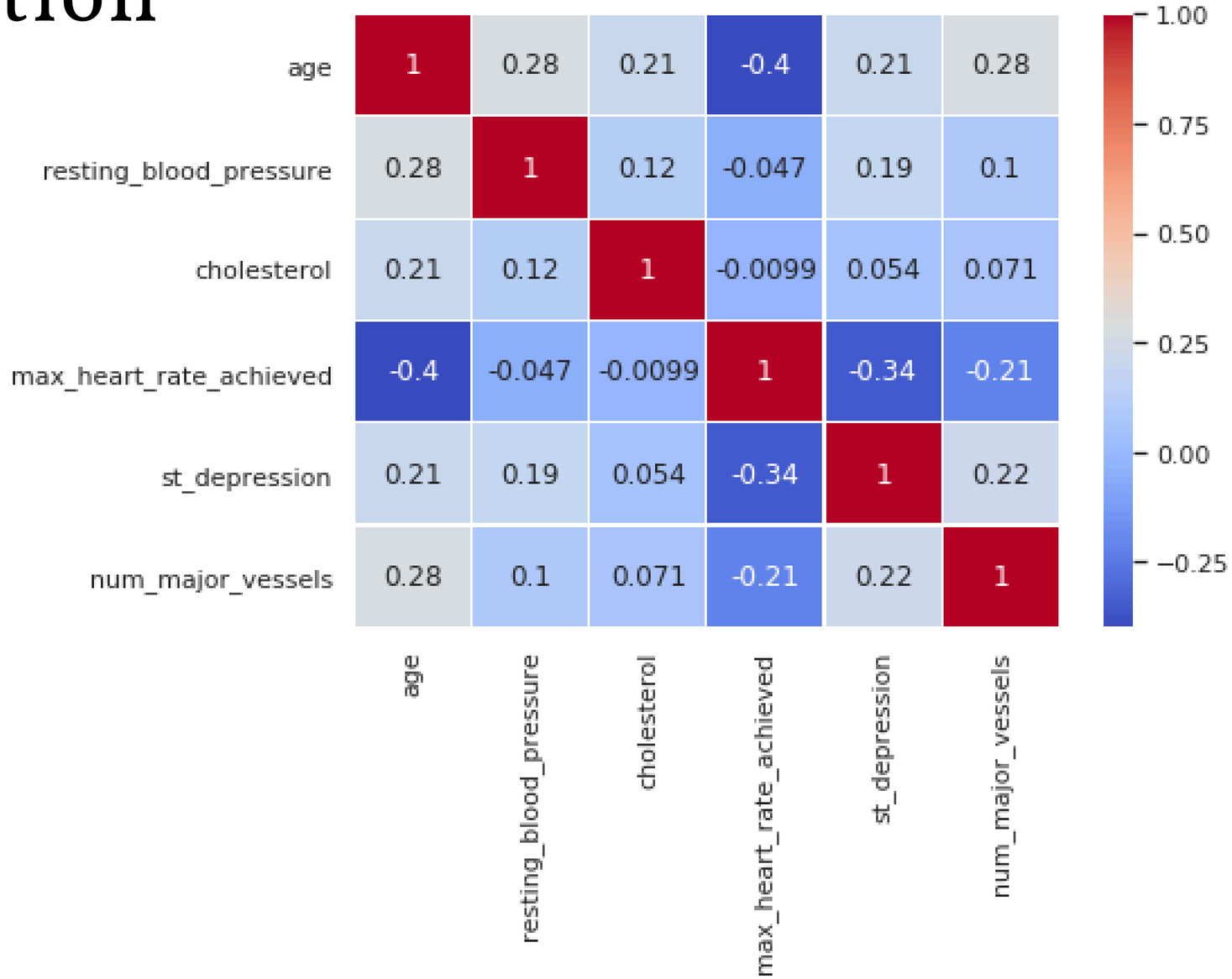




Seaborn

matplotlib

Correlation Plot



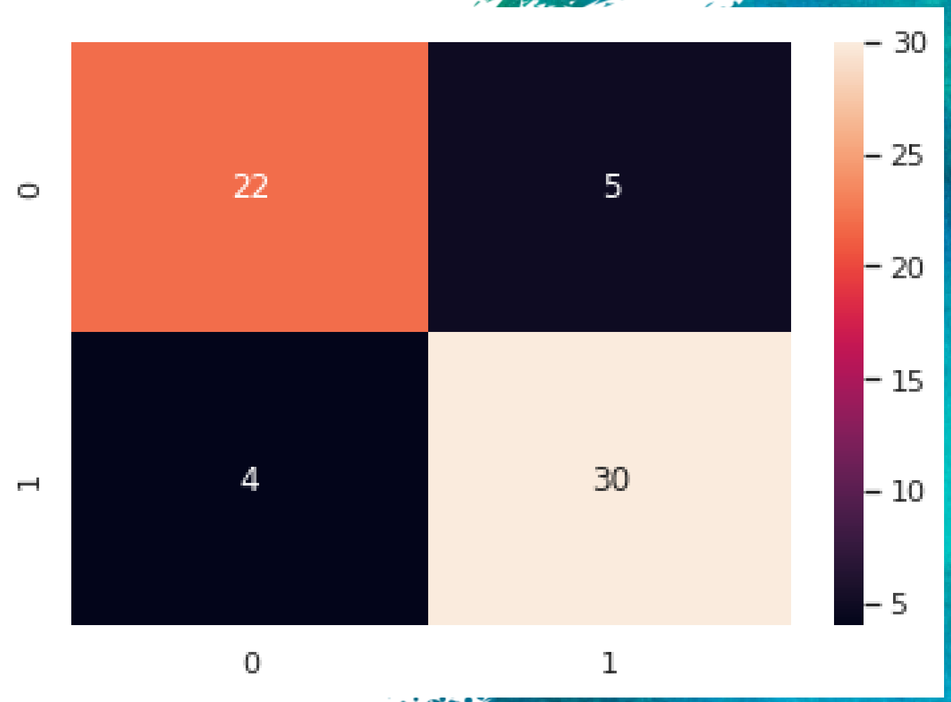
Logistic Regression

```
from sklearn.linear_model import LogisticRegression
logreg = LogisticRegression()

logreg.fit(X_train, Y_train)

y_pred_lr = logreg.predict(X_test)
print(y_pred_lr)
```

accuracy score : 85.25 %



Precision: 0.85

Recall is: 0.88

F-Score: 0.86

Random Forest

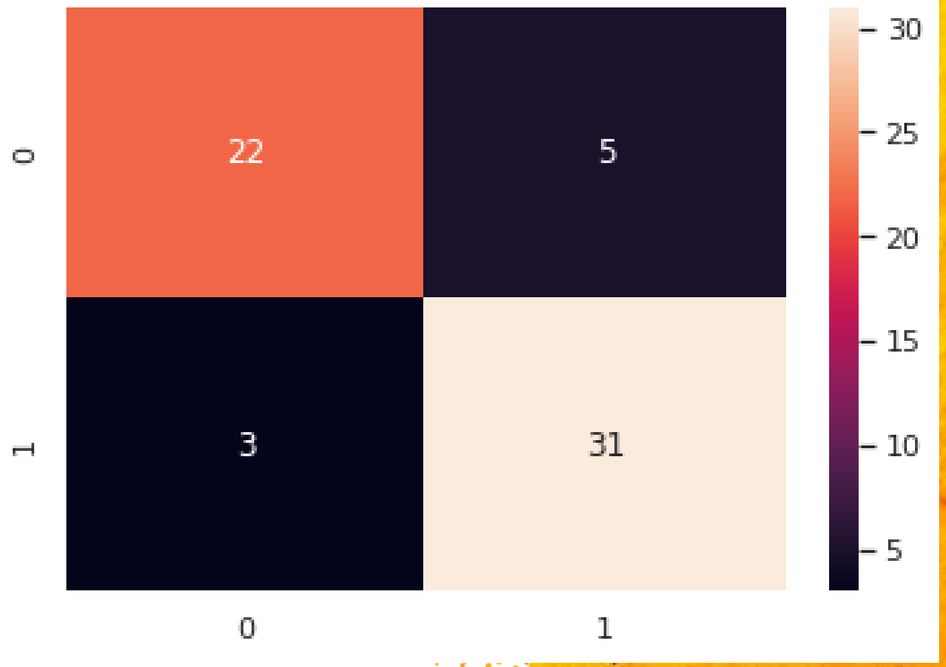
```
#Random forest with 100 trees
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators=100, random_state=0)
rf.fit(X_train, Y_train)
print("Accuracy on training set: {:.3f}".format(rf.score(X_train, Y_train)))
print("Accuracy on test set: {:.3f}".format(rf.score(X_test, Y_test)))
```

```
Accuracy on training set: 1.000
Accuracy on test set: 0.885
```

Now, let us prune the depth of trees and check the accuracy.

```
rf1 = RandomForestClassifier(max_depth=3, n_estimators=100, random_state=0)
rf1.fit(X_train, Y_train)
print("Accuracy on training set: {:.3f}".format(rf1.score(X_train, Y_train)))
print("Accuracy on test set: {:.3f}".format(rf1.score(X_test, Y_test)))
```

```
Accuracy on training set: 0.876
Accuracy on test set: 0.869
```



Precision: 0.86

Recall is: 0.91

F-Score: 0.88

Naive Bayes

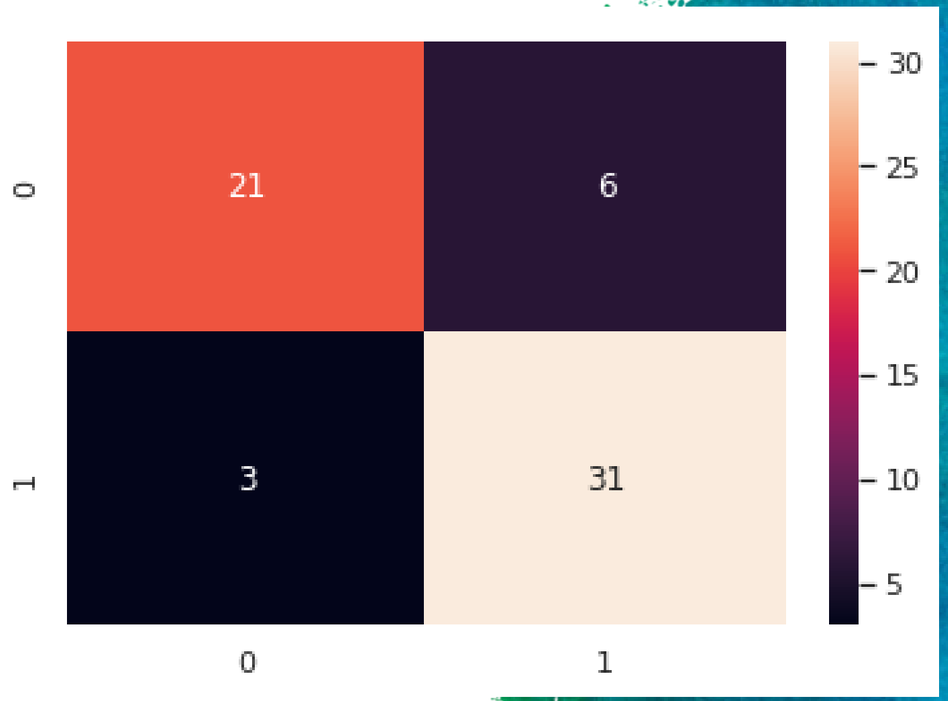
```
#Gaussian Naive Bayes
from sklearn.naive_bayes import GaussianNB
model = train_model(X_train, Y_train, X_test, Y_test, GaussianNB)
```

```
Train accuracy: 83.47%
Test accuracy: 85.25%
```

Precision: 0.83

Recall is: 0.91

F-Score: 0.87



K-Nearest Neighbor

```
from sklearn.neighbors import KNeighborsClassifier
model = train_model(X_train, Y_train, X_test, Y_test, KNeighborsClassifier)
```

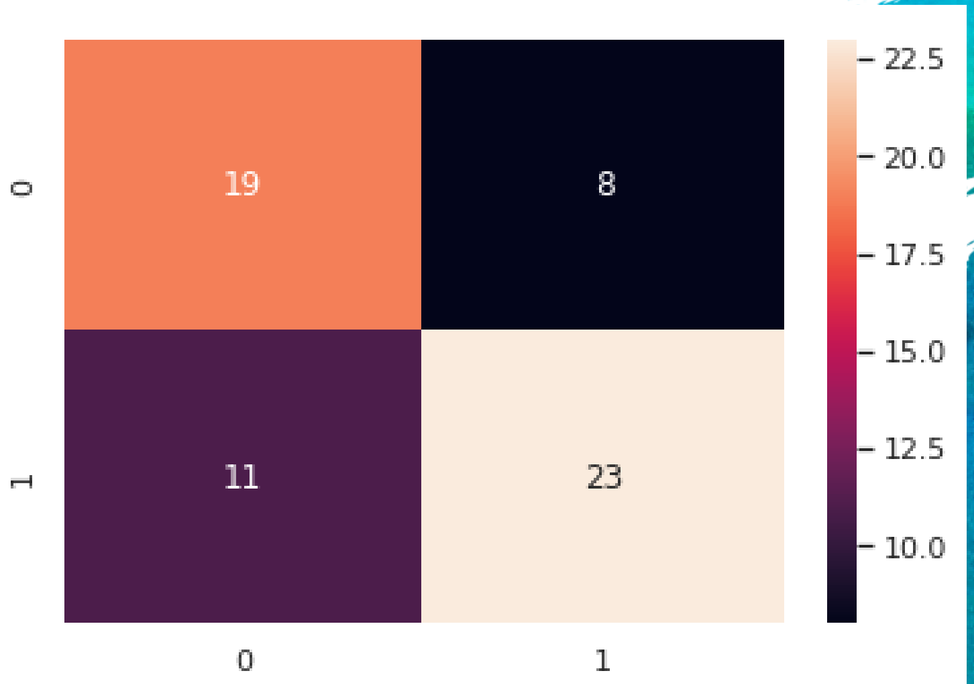
```
Train accuracy: 78.10%
Test accuracy: 63.93%
```

Let's see if KNN can perform even better by trying different 'n_neighbours' inputs.

```
# Seek optimal 'n_neighbours' parameter
for i in range(1,10):
    print("n_neigbors = "+str(i))
    train_model(X_train, Y_train, X_test, Y_test, KNeighborsClassifier, n_neighbors=i)
```

```
n_neigbors = 1
Train accuracy: 100.00%
Test accuracy: 52.46%
n_neigbors = 2
Train accuracy: 79.75%
Test accuracy: 59.02%
n_neigbors = 3
Train accuracy: 78.10%
Test accuracy: 63.93%
n_neigbors = 4
Train accuracy: 76.03%
Test accuracy: 63.93%
n_neigbors = 5
Train accuracy: 78.10%
Test accuracy: 63.93%
n_neigbors = 6
Train accuracy: 74.38%
Test accuracy: 65.57%
n_neigbors = 7
Train accuracy: 72.31%
Test accuracy: 67.21%
n_neigbors = 8
Train accuracy: 71.90%
Test accuracy: 68.85%
n_neigbors = 9
Train accuracy: 73.14%
Test accuracy: 67.21%
```

It turns out that value of n_neighbours (8) is optimal.



Precision: 0.74

Recall is: 0.67

F-Score: 0.70

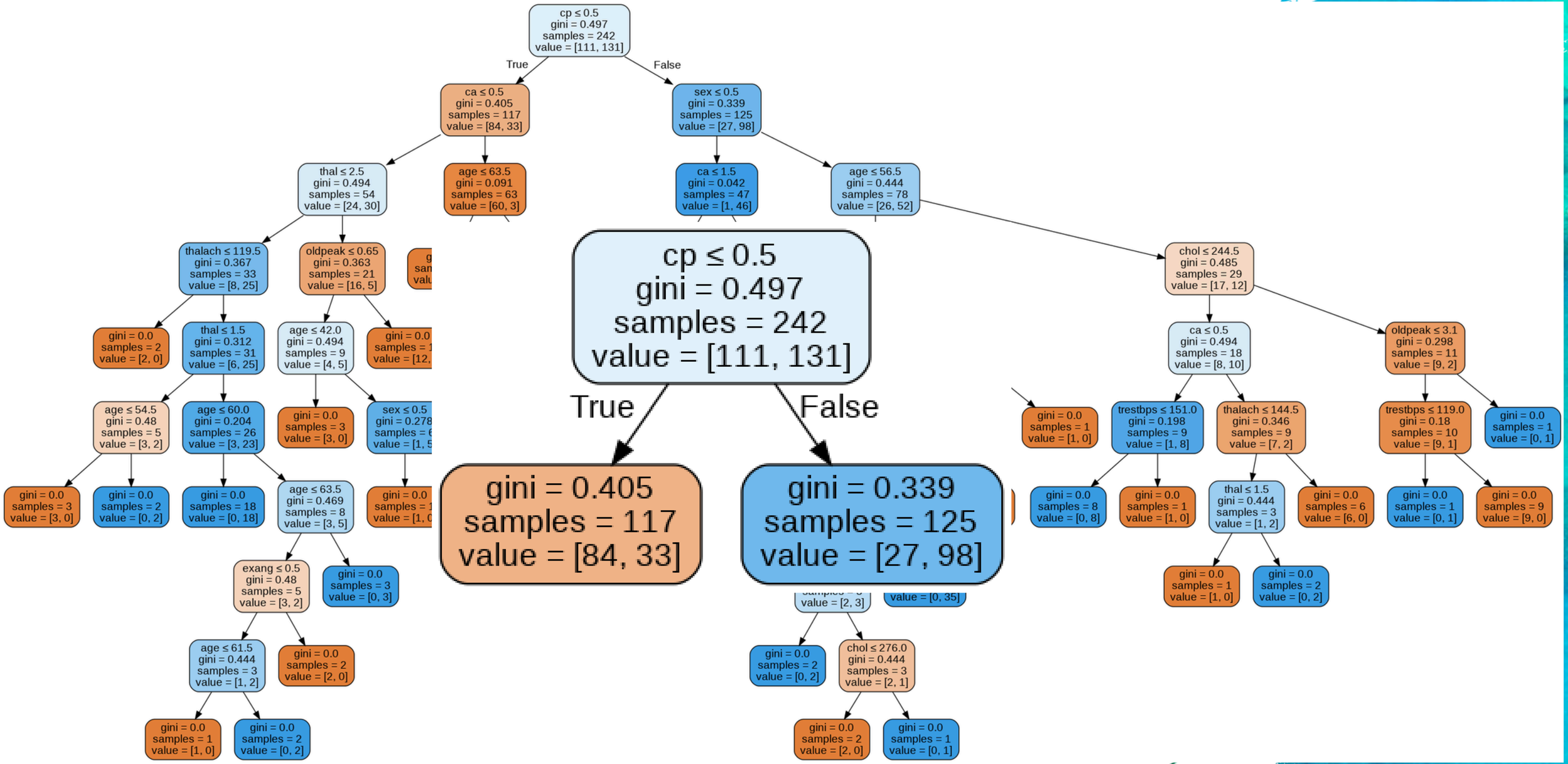
Decision Tree

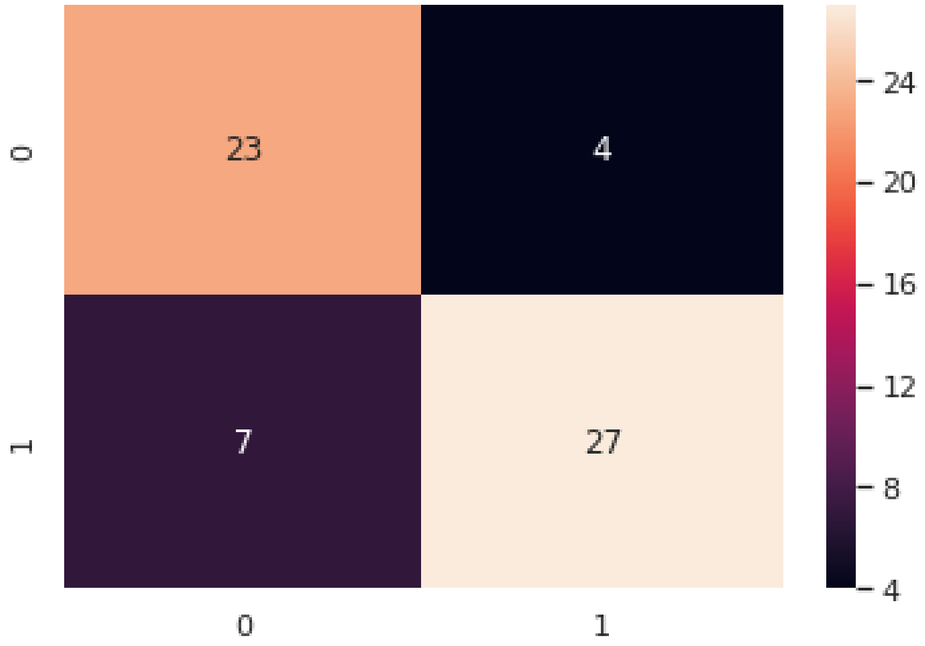
```
from sklearn.tree import DecisionTreeClassifier
tree1 = DecisionTreeClassifier(random_state=0)
tree1.fit(X_train, Y_train)
print("Accuracy on training set: {:.3f}".format(tree1.score(X_train, Y_train)))
print("Accuracy on test set: {:.3f}".format(tree1.score(X_test, Y_test)))
```

```
Accuracy on training set: 1.000
Accuracy on test set: 0.787
```

```
tree1 = DecisionTreeClassifier(max_depth=3, random_state=0)
tree1.fit(X_train, Y_train)
print("Accuracy on training set: {:.3f}".format(tree1.score(X_train, Y_train)))
print("Accuracy on test set: {:.3f}".format(tree1.score(X_test, Y_test)))
```

```
Accuracy on training set: 0.843
Accuracy on test set: 0.820
```


cp ≤ 0.5
gini = 0.497
samples = 242
value = [111, 131]
True
False
ca ≤ 0.5
gini = 0.405
samples = 117
value = [84, 33]
sex ≤ 0.5
gini = 0.339
samples = 125
value = [27, 98]
thal ≤ 2.5
gini = 0.494
samples = 54
value = [24, 30]
age ≤ 63.5
gini = 0.091
samples = 63
value = [60, 3]
ca ≤ 1.5
gini = 0.042
samples = 47
value = [1, 46]
age ≤ 56.5
gini = 0.444
samples = 78
value = [26, 52]
thalach ≤ 119.5
gini = 0.367
samples = 33
value = [8, 25]
oldpeak ≤ 0.65
gini = 0.363
samples = 21
value = [16, 5]
gini = 0.0
samples = 2
value = [2, 0]
thal ≤ 1.5
gini = 0.312
samples = 31
value = [6, 25]
age ≤ 42.0
gini = 0.494
samples = 9
value = [4, 5]
gini = 0.0
samples = 3
value = [3, 0]
sex ≤ 0.5
gini = 0.278
age ≤ 54.5
gini = 0.48
samples = 5
value = [3, 2]
age ≤ 60.0
gini = 0.204
samples = 26
value = [3, 23]
gini = 0.0
samples = 3
value = [3, 0]
gini = 0.0
samples = 2
value = [0, 2]
gini = 0.0
samples = 18
value = [0, 18]
age ≤ 63.5
gini = 0.469
samples = 8
value = [3, 5]
exang ≤ 0.5
gini = 0.48
samples = 5
value = [3, 2]
gini = 0.0
samples = 3
value = [0, 3]
age ≤ 61.5
gini = 0.444
samples = 3
value = [1, 2]
gini = 0.0
samples = 2
value = [2, 0]
gini = 0.0
samples = 1
value = [1, 0]
gini = 0.0
samples = 2
value = [0, 2]
cp ≤ 0.5
gini = 0.497
samples = 242
value = [111, 131]
True
False
gini = 0.405
samples = 117
value = [84, 33]
gini = 0.339
samples = 125
value = [27, 98]
value = [2, 3]
value = [0, 35]
gini = 0.0
samples = 2
value = [0, 2]
chol ≤ 276.0
gini = 0.444
samples = 3
value = [2, 1]
gini = 0.0
samples = 2
value = [2, 0]
gini = 0.0
samples = 1
value = [0, 1]
chol ≤ 244.5
gini = 0.485
samples = 29
value = [17, 12]
gini = 0.0
samples = 1
value = [1, 0]
ca ≤ 0.5
gini = 0.494
samples = 18
value = [8, 10]
oldpeak ≤ 3.1
gini = 0.298
samples = 11
value = [9, 2]
trestbps ≤ 151.0
gini = 0.198
samples = 9
value = [1, 8]
thalach ≤ 144.5
gini = 0.346
samples = 9
value = [7, 2]
trestbps ≤ 119.0
gini = 0.18
samples = 10
value = [9, 1]
gini = 0.0
samples = 1
value = [0, 1]
gini = 0.0
samples = 8
value = [0, 8]
gini = 0.0
samples = 1
value = [1, 0]
thal ≤ 1.5
gini = 0.444
samples = 3
value = [1, 2]
gini = 0.0
samples = 6
value = [6, 0]
gini = 0.0
samples = 1
value = [0, 1]
gini = 0.0
samples = 9
value = [9, 0]
gini = 0.0
samples = 1
value = [1, 0]
gini = 0.0
samples = 2
value = [0, 2]

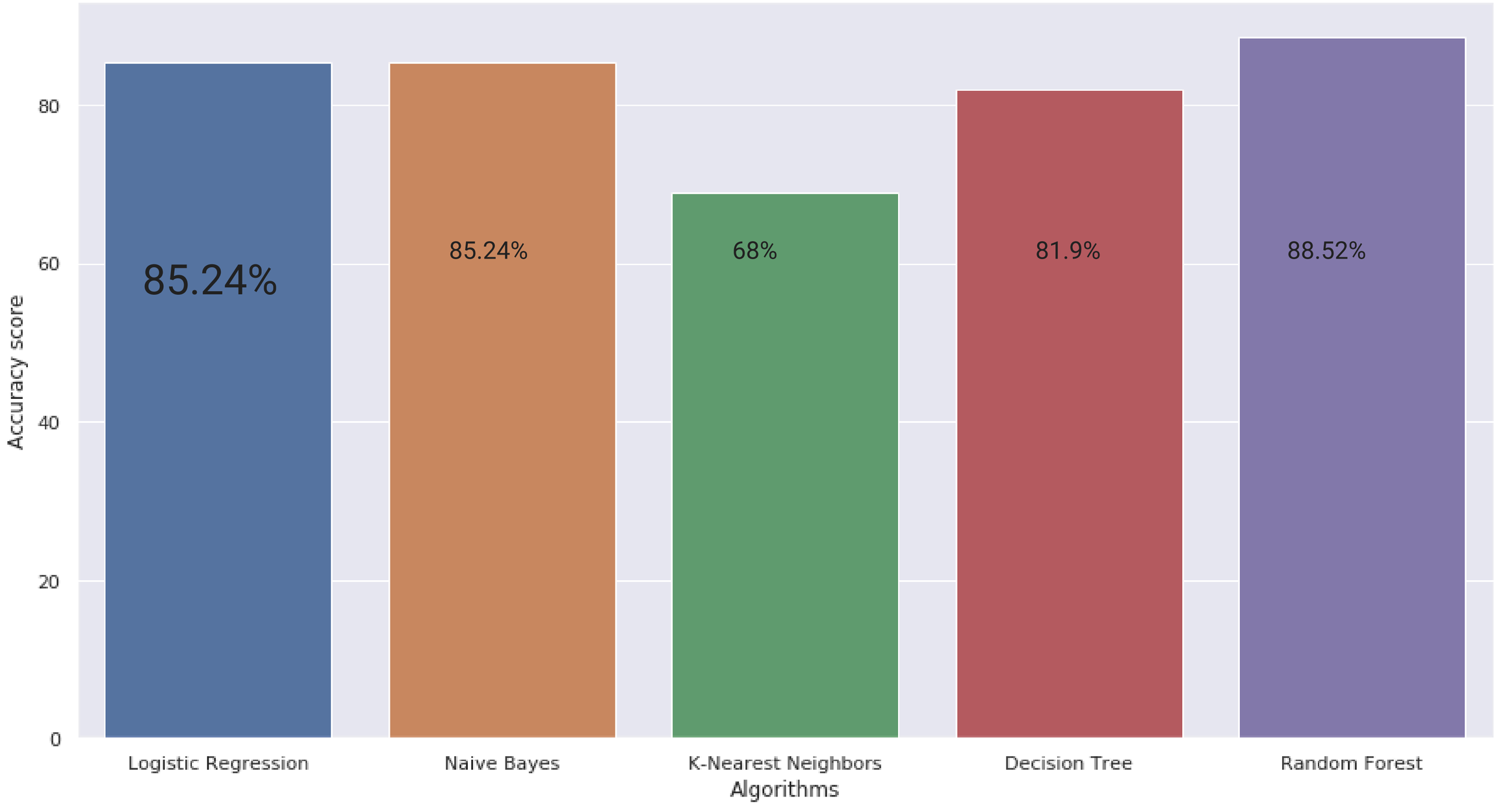


Precision: 0.87

Recall is: 0.79

F-Score: 0.83

Results


Accuracy score
0
20
40
60
80
85.24%
85.24%
68%
81.9%
88.52%
Logistic Regression
Naive Bayes
K-Nearest Neighbors
Decision Tree
Random Forest
Algorithms


thank you